

15.6

- (a) The OLS estimates for 1987 and 1988 are similar. However, these separate OLS regressions do not control for observed individual heterogeneity, so the estimates may suffer from omitted variable bias.
- (b) The panel model decomposes the error term into $u_i + e_{it}$. u_i captures time-invariant individual heterogeneity, e_{it} is the idiosyncratic error. Compared with OLS, the panel model explicitly accounts for individual differences.
- (c) The FE estimates differ from OLS, especially for EXPER and SOUTH. EXPER becomes much smaller under FE, suggesting that OLS may overstate the effect of experience because of omitted individual characteristics. FE remove u_i , so it reduces omitted variable bias from time-invariant individual factors.

(d) $df_1 = 716 - 1 = 715$

$$df_2 = 1432 - 716 - 4 = 712$$

$$F_{0.99, 715, 712} = 1.19$$

$11.68 > 6.19$, we reject H_0 and conclude that there are individual differences in the fixed effects.

- (e) Column (3) uses conventional FE standard errors, while column (4) uses cluster-robust standard errors. Cluster-robust standard errors allow heteroskedasticity and within-individual serial correlation. The robust standard errors are generally larger especially for SOUTH, suggesting that conventional standard errors may underestimate uncertainty.

(f) $H_0: \beta_{FE,k} = \beta_{RE,k}$

$$t = \frac{b_{FE,k} - b_{RE,k}}{\sqrt{se(b_{FE,k})^2 - se(b_{RE,k})^2}}$$

$$t_{EXPER} = \frac{0.0575 - 0.0988}{\sqrt{0.033^2 - 0.022^2}} = -1.67$$

$$t_{EXPER^2} = \frac{-0.0012 - (-0.0023)}{\sqrt{0.0011^2 - 0.0007^2}} = 1.296$$

$$t_{SOUTH} = \frac{-0.3261 - (-0.2326)}{\sqrt{0.1258^2 - 0.0817^2}} = -0.77$$

$$t_{UNION} = \frac{0.0822 - 0.1027}{\sqrt{0.0312^2 - 0.0245^2}} = -1.06$$

$$|t_{EXPER}| = 1.67$$

$\Rightarrow$ EXPER is the coefficient that differs most from the fixed effects estimate.

At the 5% level, $|t| = 1.96$, $1.67 < 1.96$, the largest difference is not significant, so we do not reject H_0 .

15.20

(a) Call:
lm(formula = eq_base, data = star)

Residuals:

	Min.	1Q	Median	3Q	Max.
	-107.220	-20.214	-3.935	14.339	185.956

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	437.76425	1.34622	325.180	< 2e-16 ***
small	5.82282	0.98933	5.886	4.19e-09 ***
aide	0.81784	0.95299	0.858	0.391
tchexper	0.49247	0.06956	7.080	1.61e-12 ***
boy	-6.15642	0.79613	-7.733	1.23e-14 ***
white_asian	3.90581	0.95361	4.096	4.26e-05 ***
freelunch	-14.77134	0.89025	-16.592	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 5759 degrees of freedom
(因為不存在, 20 個觀察量被刪除了)

Multiple R-squared: 0.09685, Adjusted R-squared: 0.09591
F-statistic: 102.9 on 6 and 5759 DF, p-value: < 2.2e-16

Small classes have a positive and significant effect on reading scores.

Aide is positive but not significant.

Teacher experience has a positive effect.

Boys score lower, White and Asian students score higher and students receiving free lunch score substantially lower.

(b) Call:
plm(formula = eq_base, data = pdata, model = "within")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:

	Min.	1st Qu.	Median	3rd Qu.	Max.
	-102.6381	-16.7834	-2.8473	12.7591	198.4169

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
small	6.490231	0.912962	7.1090	1.313e-12 ***
aide	0.996087	0.881693	1.1297	0.2586
tchexper	0.285567	0.070845	4.0309	5.629e-05 ***
boy	-5.455941	0.727589	-7.4987	7.440e-14 ***
white_asian	8.028019	1.535656	5.2277	1.777e-07 ***
freelunch	-14.593572	0.880006	-16.5835	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 4628000
Residual Sum of Squares: 4268900
R-Squared: 0.077592
Adj. R-Squared: 0.063954
F-statistic: 79.6471 on 6 and 5681 DF, p-value: < 2.22e-16

Small class effect remains positive and significant, increasing from about 5.82 to 6.49.

Teacher aides remains insignificant.

Teacher experiences became smaller, suggesting that OLS effect may reflect school-level differences.

Free-lunch effect remains strongly negative.

(c) Analysis of Variance Table

Model 1: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
Res. Df 5759 Sum of Sq 5247584

Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch + factor(schid)
Res. Df 5681 Sum of Sq 4268894

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$H_0: \beta_{11} = \beta_{12} = \dots = \beta_{1N}$ vs H_1 : the β_{1i} are not all equal

$F = 16.678$, $p\text{-value} < 0.001$. We reject H_0 and conclude that school fixed effects are significant.

If school effects are uncorrelated with the regressors, the inclusion of fixed effects should have little influence on the remaining coefficient estimates.

(d) Call:
plm(formula = eq_base, data = pdata, model = "random")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Effects:

	var	std.dev	share
idiosyncratic	751.43	27.41	0.829
individual	155.31	12.46	0.171

theta:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.6470	0.7225	0.7523	0.7541	0.7831	0.8153

Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-97.4830	-17.2361	-3.2822	0.0372	12.8027	192.3460

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	436.126774	2.064782	211.2217	< 2.2e-16 ***
small	6.458722	0.912548	7.0777	1.466e-12 ***
aide	0.992146	0.881159	1.1260	0.2602
tchexper	0.302679	0.070292	4.3060	1.662e-05 ***
boy	-5.512081	0.727639	-7.5753	3.583e-14 ***
white_asian	7.350477	1.431376	5.1353	2.818e-07 ***
freelunch	-14.584332	0.874676	-16.6740	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 6158000
Residual Sum of Squares: 4332100
R-Squared: 0.29655
Adj. R-Squared: 0.29582
Chisq: 493.205 on 6 DF, p-value: < 2.22e-16

RE estimator is similar in OLS and fixed effects.

The LM test $p\text{-value} < 0.05$, so we reject H_0 .

Significant school random effects exist, and the random effects model is preferred over

pooled OLS.

```
> plmtest(pooled_plm, effect = "individual", type = "bp")
```

Lagrange Multiplier Test - (Breusch-Pagan)

```
data: eq_base
chisq = 6677.4, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects
```

(e) small aide tchexper white_asian freelunch boy
1.14600764 0.12843803 -1.93771666 1.21807432 -0.09555102 NaN

Hausman Test

```
data: eq_base
chisq = 13.809, df = 6, p-value = 0.03184
alternative hypothesis: one model is inconsistent
```

At the 5% level, there is no significant differences between the FE and RE estimates.

but there is an undefined result for BOY.

$p\text{-value} < 0.05$, so we reject H_0 , so endogeneity exists, RE estimator should not be used.

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept)	459.462989	20.529888	22.3802	< 2.2e-16	***
small	6.637460	0.922068	7.1985	6.090e-13	***
aide	1.157620	0.889542	1.3014	0.1931	
tchexper	0.289286	0.071754	4.0316	5.539e-05	***
boy	-5.386109	0.735063	-7.3274	2.346e-13	***
white_asian	8.081423	1.550155	5.2133	1.855e-07	***
freelunch	-14.699025	0.892109	-16.4767	< 2.2e-16	***
m_small	-18.410060	22.273923	-0.8265	0.4085	
m_aide	16.811358	20.793685	0.8085	0.4188	
m_tchexper	1.006007	0.625690	1.6078	0.1079	
m_boy	-53.353521	25.221654	-2.1154	0.0344	*
m_white_asian	-6.648191	6.320012	-1.0519	0.2928	
m_freelunch	-3.318853	8.779553	-0.3780	0.7054	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 6007200
Residual Sum of Squares: 4281300
R-Squared: 0.28737
Adj. R-Squared: 0.28586
Chisq: 500.306 on 12 DF, p-value: < 2.22e-16

Linear hypothesis test:

m_small = 0
m_aide = 0
m_tchexper = 0
m_boy = 0
m_white_asian = 0
m_freelunch = 0

Model 1: restricted model

Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
m_small + m_aide + m_tchexper + m_boy + m_white_asian + m_freelunch

Note: Coefficient covariance matrix supplied.

	Res.Df	Df	Chisq	Pr(>Chisq)
1	5674			
2	5668	6	17.5	0.007611 **

$$\chi^2 = 17.5 > \chi^2_{0.95, 6} = 12.59$$

p-value < 0.05, reject H_0 .

Unobserved school effects are correlated with the regressors, RE estimator

should not be used.